

REMNANT FIELDWORKS INC.

Inheritance Math Framework (IMF): Formal Foundations of the Coherent Inheritance Framework

v1.2.1 Release Integrity Reconciliation (patch)
Phase 1 Candidate Framework for Open Review

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"This document presents a candidate phenomenological modeling framework. All formal results are conditional on stated assumptions. No universal theorems about physical reality are claimed. The framework is offered for open critique, testing, and falsification."

Abstract

The Inheritance Math Framework (IMF) is the formal mathematical core of the Coherent Inheritance Framework (CIF) — a candidate phenomenological modeling vocabulary for studying how state, structure, and constraints propagate through inheritance chains across diverse system types.

This v1.2.1 patch release applies additional targeted corrections to v1.2, addressing operator-order and model-description issues identified in subsequent review. The coherence balance postulate and all simulation code are now unified under a single equation, $\dot{C} = -\alpha\bar{D} + \beta F + \gamma Ru - \delta C$; channel fidelity is renamed F (positive coefficient β) throughout, retiring the $\bar{W}$ notation that carried an incorrect sign in the simulation code. Proposition 4.3 is reformulated to measure actual-versus-ideal trajectory deviation $E_k = d(\rho_k^{\text{actual}}, \rho_k^{\text{ideal}})$ rather than deviation from the initial state; the prior formulation incorrectly treated expected ideal evolution as error. Proposition 4.2 is correctly scoped as an instantaneous boundary non-exit condition rather than a global viability guarantee. Proposition 4.5 adds the required drift-rate lower bound assumption $v_{\min} > 0$ that makes the finite warning-time estimate valid. An unsupported claim about error frequency is removed. Adaptive capacity is resolved as a domain-specific coefficient γ rather than an unnamed separate state variable. Language throughout is corrected to read "illustrated" rather than "validated" through simulations, and a claim of preliminary cross-domain evidence is replaced by the accurate "preliminary cross-domain conceptual correspondence."

The simulation suite has been rebuilt in full to implement Model Postulate 4.1 exactly. Four lane instantiations, five propositions, and five empirical hypotheses are preserved. The framework remains a Phase 1 candidate, offered for open review, critique, and falsification.

Keywords: coherence, inheritance, drift, channel fidelity, resonance, adaptive capacity, quantum information, organizational dynamics, epistemic classification, actual-vs-ideal error

Version History

Version	Date	Change Summary
v1.2.1	2026	Release Integrity Reconciliation — patch:
		• Prop 4.3 proof fixed: Corrected operator order to $N(J(\rho))$ — perturbation applied <i>after</i> channel — yielding the correct split $\varepsilon_{k+1} + L_{k+1} \cdot E_k$. Prior v1.2 had $J(N(\rho))$ which gives $L(E_k + \varepsilon)$, mismatching the stated recurrence.
		• Projected-model remark: Replaced parameter-domain condition with accurate description: simulations implement $\Pi_{[0,1]}[C_n + \Delta t \cdot f(C_n)]$; negative raw equilibria represent lower-boundary pressure, not violations.
		• State-vs-trajectory error distinction added: Prop 4.3 bounds state-at-landmark error E_k^{state} ; trajectory-space error is a distinct concept deferred to future work.
		• Figure 1 description replaced: Was: constant disruption levels; now: growing disruption, delayed stewardship at $t = 15$, resonance-matched from $t = 0$.
		• Figure 6 description replaced: Was: running warning-time estimate, $C_\theta = 0.5$; now: mean coherence/drift panels, AUC bars, $C_{\min} = 0.25$, post-collapse exclusion.
		• Convergence table removed: Replaced with reference to standalone <code>solver_convergence_check.py</code> and <code>solver_convergence_results.csv</code> .
		• Figure 2 regime labels updated: From "High/Medium/Low Fidelity" to "Near-unity/Moderate/Strong contractivity"; added note on floor-ordering relationship.
		• Figure 4 endpoint annotations fixed: Show $C(60)$ (final simulated value), not $C^*(0)$ (initial equilibrium formula).

Release Integrity Reconciliation:

- **Prop 4.3 fixed:** Added ε_k perturbation term to recurrence; updated closed-form bound to $E_K \leq L^K E_0 + \varepsilon(1 - L^K)/(1 - L)$; established stable error floor $E_\infty = \varepsilon/(1 - L)$.
- **Hypothesis 7.5 updated:** from "error decreases" to "error converges to positive stable floor".
- **Postulate 4.1 extended:** Added parameter-domain condition ensuring $C^* \in [0, 1]$.
- **Prop 4.6 fixed:** Explicit opening hypothesis $C^* \geq C_\theta$ (equivalently $-\alpha\bar{D} + \beta F + \gamma Ru \geq \delta C_\theta$) added.
- **Assumption 2.9 replaced:** Statistical independence removed; replaced with narrowed identification note.
- **Control-lane attribution fixed:** Fidelity result now correctly attributed to βF equilibrium term, not Proposition 4.4.
- **Consistency claim softened:** "All formal results are internally consistent" replaced by "presented for further consistency review".
- **Section 8 fixed:** Each figure's mathematical basis accurately classified (ODE, discrete recurrence, categorical, auxiliary model).
- **Name correction:** "Integrated Maintenance Framework" $\rightarrow$ "Inheritance Math Framework (IMF)" throughout.
- **References contradiction fixed:** Section now correctly states framework draws on, rather than ignores, cited literature.
- **Figure numbering standardized:** Figure 1–6 throughout (not Figure 8.X).
- **Figure 5 caption updated:** Categorical-label nature of alignment mapping explicitly stated.

v1.2 2026

v1.0 Aug 2026

Initial public review draft. Core balance law presented as derived theorem. Six results labeled as theorems. Channel fidelity referenced to ρ^* . Jensen–Shannon divergence used directly (not metric form). Resonance defined as function of drift. No dimensional normalization.

v1.1 Aug 2026

Mathematical correction: postulate reclassification (Balance Law $\rightarrow$ Model Postulate 4.1); JS metric corrected to $\sqrt{D_{JS}}$; resonance redefined as intervention alignment independent of drift; all scalars normalized to $[0,1]$; five propositions rebuilt; ExecutionProof mapping revised (Ev = source validity; Merkle limitation noted); epistemic tags added throughout; Section 0 notation table added.

v1.1.1 Aug
2026

Second targeted correction pass:

- **ODE unification:** Postulate 4.1 and simulation code now both implement $\dot{C} = -\alpha\bar{D} + \beta F + \gamma Ru - \delta C$ exactly.
- **$\bar{W} \rightarrow F$ rename:** Channel fidelity F with positive coefficient β replaces $\bar{W}$, which carried an incorrect sign in v1.1 code.
- **Prop 4.3 reformulated:** Error measure changed from $d(\rho_K, \rho_0)$ to $E_k = d(\rho_k^{\text{actual}}, \rho_k^{\text{ideal}})$; contractivity constants L_k added; frequency claim removed.
- **Prop 4.2 rescoped:** Renamed instantaneous boundary non-exit condition; global viability requires the condition to hold at all boundary times (stated explicitly).
- **Prop 4.5 extended:** Drift-rate lower bound assumption $v_{\min} > 0$ added for finite warning-time bound.
- **Adaptive capacity resolved:** Identified as coefficient γ ; no separate state variable.
- **Language corrections:** "illustrated" not "validated"; "conceptual correspondence" not "preliminary evidence".
- **Simulation suite rebuilt in full** under corrected postulate.

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0. Notation and Conventions

Symbols marked † are new or redefined in v1.1.1; double-dagger ‡ marks changes in v1.2. The symbol $\bar{W}$ (used in v1.0 and v1.1 for channel fidelity/waste) is retired and replaced throughout by F .

Symbol	Meaning	Range Units	/ Defined at
$\mathcal{S}$	State space (compact metric space)	—	Assumption 2.1
$\rho(t), \rho^*$	System state; reference (coherent) state	$\mathcal{S}$	Assumptions 2.1–2.2
$F = (\mathcal{S}, T, I)$	Field structure: state space, transition law, inheritance channel	—	§1.5
J	Ideal inheritance channel	$\mathcal{S}_P \rightarrow \mathcal{S}_C$	Definition 3.1
N_k	Perturbation operator at step k (preserves state space structure)	CPTP / stochastic kernel	Definition 3.6
$d(\cdot, \cdot)$	Domain-appropriate metric	$\mathbb{R}_{\geq 0}$	Assumption 2.1
$\bar{D}(t)$	Normalized drift $\in [0, 1]$ (negative effect on C)	$[0, 1]$	Definition 3.2
$F(t)^\dagger$	Channel fidelity $\in [0, 1]$ (positive effect on C, coefficient β)	$[0, 1]$	Assumption 2.11 / Def 3.1
$C(t)$	Coherence functional $\in [0, 1]$	$[0, 1]$	Definition 3.5
$C_{\min}$	Viability floor	$(0, 1)$	Definition 3.7
$R(t)$	Resonance — intervention alignment $\in [0, 1]$	$[0, 1]$	Assumption 2.10
$\bar{u}(t)$	Normalized stewardship input $\in [0, 1]$	$[0, 1]$	Assumption 2.6
$\alpha, \beta, \gamma, \delta$	Rate coefficients: drift loss, fidelity gain, resonance gain, natural decay	$T^{-1}, \geq 0$	Postulate 4.1
$E = \{\rho : h_i(\rho) \leq 0\}$	Non-harm (admissible) envelope; h_i are constraint functions	subset of $\mathcal{S}$	Definition 3.7
$\pi = (A, \text{Ev}, K, \text{Rec})$	Proof state: Authority, Evidence, Constraints, Record	—	Definition 3.7
F_I	Inheritance channel fidelity (relative to J)	$[0, 1]$	Definition 3.1
$E_k^\dagger$	Actual-vs-ideal state-at-landmark error at generation k	≥ 0	Definition 3.8
ϵ_k	Per-generation perturbation bound: $d(N_k(x), x) \leq \epsilon_k$	≥ 0	Proposition 4.3

L_k †	Contractivity constant of ideal channel J_k	$[0, 1]$	Proposition 4.3
φ †	Monotone coherence-from-drift function: $C = \varphi(\overline{D}), \varphi' \leq -m < 0$	$\mathbb{R}_{\geq 0} \rightarrow [0, 1]$	Proposition 4.5
$v_{\min}$ †	Minimum drift growth rate (lower bound for finite time estimate)	$T^{-1}, > 0$	Proposition 4.5
AUC †	Area under ROC curve (landmark-time methodology)	$[0, 1]$	Section 8, Figure 6
D_{JS}	Jensen–Shannon divergence; metric form: $\sqrt{D_{\text{JS}}}$	$[0, \ln 2]$	Definition 3.6
$\overline{W}$ (retired)	Retired v1.0/v1.1 symbol; replaced by F throughout	—	See v1.1 changelog

† Symbol new or redefined in v1.1.1. ‡ Symbol updated in v1.2.

1. Introduction

1.1 Motivation

Across radically different scientific domains — quantum information theory, machine learning, control engineering, and organizational management — a common phenomenology recurs: a structured system defined by an initial coherent state tends to drift from that state over time unless actively maintained. The quantum computer's entangled register decoheres under environmental noise; the trained neural network's predictions degrade under distributional shift; the factory's calibrated production process drifts from specification; the organization's strategic alignment erodes under personnel turnover and changing priorities.

These phenomena share a mathematical skeleton: a state space, a target configuration, a drift process pushing the system away, a transmission channel of variable quality, and optional interventions that can slow or reverse the drift. The Inheritance Math Framework (IMF) is an attempt to make that skeleton explicit, domain-agnostic, and mathematically tractable as a candidate phenomenological vocabulary — a Phase 1 framework offered for open critique and falsification.

1.2 Scope

IMF provides formal mathematical language for the Coherent Inheritance Framework (CIF). It does not claim to derive new fundamental results in quantum information, control theory, or information theory. Its value is architectural: a shared vocabulary that enables comparable questions to be asked across domains, and a structured protocol for experimental validation. The framework is domain-agnostic; its quantitative predictions in any specific domain are subject to the empirical hypotheses of Section 7.

1.3 What This Document Does and Does Not Claim

This document claims: (a) the formal structure has undergone three correction passes and is presented for further consistency review — as documented in the version history, each pass has identified and corrected genuine structural issues; (b) the four lane instantiations of Section 5 are plausible mappings of the abstract framework to concrete domain objects; (c) the simulations of Section 8 correctly illustrate the model's postulated behavior for their respective mathematical objects. This document does *not* claim that any real system follows the coherence balance law, that the lane analogies hold quantitatively, or that the simulations constitute empirical evidence from real systems.

1.4 Relationship to Prior Versions

Version 1.0 presented six alleged "theorems," several of which were definitional statements or carried invalid inferences. Version 1.1 reclassified the core equation as a Model Postulate, repaired the Jensen–Shannon metric, separated resonance from drift, introduced normalized variables, and rebuilt five propositions with explicit assumptions. Version 1.1.1 corrects the remaining issues identified in a second adversarial review:

- **ODE mismatch:** The v1.1 simulation code implemented a different algebraic form from the whitepaper's Postulate 4.1 ($k_R R + k_u \bar{u} + k_{Ru} Ru - k_D \bar{D} - k_W \bar{W}$ vs. $-\alpha \bar{D} + \beta \bar{W} + \gamma Ru - \delta C$), and treated $\bar{W}$ as a waste/negative term. Both forms are now unified into the single canonical equation $\dot{C} = -\alpha \bar{D} + \beta F + \gamma Ru - \delta C$.
- **Sign error / notation:** $\bar{W}$ carried an incorrect negative coefficient in code. Retired. Channel fidelity is now F with positive coefficient β .
- **Proposition 4.3:** Error measured against ρ_0 (initial state) treated expected ideal evolution as error. Reformulated around actual-vs-ideal state-at-landmark error $E_k = d(\rho_k^{\text{actual}}, \rho_k^{\text{ideal}})$ with contractivity constants L_k .
- **Proposition 4.2:** Stated as global viability guarantee. A derivative condition at one instant does not guarantee future viability. Correctly scoped as instantaneous boundary non-exit condition.
- **Proposition 4.5:** Did not produce a finite warning-time estimate without an additional assumption. Drift-rate lower bound $v_{\min} > 0$ added for the time bound.
- **Frequency claim:** "Frequent small errors more damaging than rare large errors" removed — does not follow from the triangle bound.
- **Adaptive capacity:** Named in prior abstracts as a fourth interacting quantity but absorbed into γ without explanation. Resolved: γ is the domain-specific adaptive capacity coefficient.
- **Language:** "Validated qualitatively through simulations" → "illustrated through simulations"; "preliminary cross-domain evidence" → "preliminary cross-domain conceptual correspondence."

1.5 Forest–Trunks–Branches–Leaves Hierarchy

IMF organizes domain knowledge in a four-tier hierarchy. The *Forest* is the domain itself — quantum, AI, control, or organizational — the context in which coherence matters. *Trunks* are persistent structural patterns: the target state ρ^* , the admissible set $\mathcal{A}$, the ideal channel J . *Branches* are operational subsystems — the four lanes (Section 5) — each applying the Balance Law with domain-specific instantiations. *Leaves* are individual transmission events where error accumulation begins.

The field structure $F = (\mathcal{S}, T, I)$ consists of: the state space $\mathcal{S}$, an autonomous transition law $T : \mathcal{S} \times \mathbb{R}_{\geq 0} \rightarrow \mathcal{S}$ governing unperturbed evolution, and an inheritance channel $I : \mathcal{S}_P \rightarrow \mathcal{S}_C$ mapping parent states to child states. The ideal channel J defines what inheritance *would* produce without perturbation. The ProofRecord system (Section 6) records events at the leaf level and aggregates them upward toward trunk-level viability assessments.

1.6 Paper Organization

Section 0 fixes notation. Sections 2–3 establish foundational assumptions and core definitions. Section 4 presents the model postulate and five propositions with proofs. Section 5 gives lane-by-lane instantiations. Section 6 maps to ExecutionProof / ProofRecord. Section 7 states five empirical hypotheses with experimental protocols. Section 8 presents rebuilt numerical simulations. Section 9 provides an honest epistemic assessment including adjacency comparisons. Sections 10–11 describe the RF Inc. connection and conclude.

ASSUMPTION 2.1 Bounded Coherence State. The coherence variable $C(t) \in [0, C_{\max}]$ for some finite $C_{\max} > 0$. Coherence is non-negative and bounded above by a domain-specific ceiling. For analysis, we often normalise so that $C_{\max} = 1$.

ASSUMPTION 2.2 Continuous Differentiability. The function $C(t)$ is continuously differentiable on its domain of definition. Discrete-time implementations approximate this with sufficiently small step sizes.

ASSUMPTION 2.3 Well-Defined Disruption Magnitude. For any system trajectory, the disruption $D(t) \geq 0$ is measurable and its time-average $\bar{D}$ is well-defined. Disruption is treated as an exogenous or semi-exogenous driving signal.

ASSUMPTION 2.4 Well-Defined Control Input. The control input $u(t) \in [0, 1]$ is measurable and its mean $\bar{u}$ is well-defined over the time horizon of interest. Control represents the normalised effort or resource allocation applied to coherence maintenance.

ASSUMPTION 2.5 Resonance Measurability. The resonance metric $R \in [0, 1]$ is a well-defined, observable or estimable quantity for each system instance. R captures the degree to which control effort and system structure are mutually aligned.

ASSUMPTION 2.6 Constant Coefficients. The parameters $\alpha, \beta, \gamma, \delta > 0$ are constant over the time horizon under analysis. Time-varying extensions are permitted but require re-derivation of propositions.

ASSUMPTION 2.7 Linear Superposition of Influences. The instantaneous rate of change of coherence is a linear sum of driving terms. No higher-order interaction terms are included in the base model. This assumption may be relaxed in extended formulations.

ASSUMPTION 2.8 Decay Proportionality. The passive decay of coherence is proportional to its current level, with rate coefficient $\delta > 0$. This reflects the empirical observation that coherence erodes faster when it is higher, absent active maintenance.

NOTE 2.9 On Identification of Input Quantities.  **Correction from v1.0/v1.1:** Replaces v1.1.1 Assumption 2.9 (statistical independence). Statistical independence was an over-strong foundational requirement; it is here downgraded to a modelling note. The four input quantities $\bar{D}$, F , R , and $\bar{u}$ may be empirically correlated. The ODE of Postulate 4.1 does not require statistical independence for its mathematical formulation — it is a deterministic (or stochastic) differential equation in state $C(t)$. In empirical estimation contexts, one may impose a working-independence assumption for parameter identification; this is a modelling choice, not a foundational requirement. The simulation suite intentionally couples these variables (e.g., a stewardship intervention simultaneously raises $\bar{u}$, R , and F while reducing $\bar{D}$) to reflect realistic dynamics.

ASSUMPTION 2.10 Resonance Alignment. When $R = 0$ the resonance term contributes nothing to $\dot{C}$ regardless of control effort $\bar{u}$. Maximal amplification of control input occurs at $R = 1$. This multiplicative structure captures the intuition that control effort is wasted without structural alignment.

ASSUMPTION 2.11 Channel Fidelity as a Positive Driver. The channel fidelity $F \in [0, 1]$ enters the ODE with a strictly positive coefficient $\beta > 0$. Higher fidelity of information or interaction channels between system components increases the coherence growth rate. This corrects the v1.0/v1.1 convention in which the analogous variable $\bar{W}$ had ambiguous sign; see §1.4 and the Change Log.

Together these eleven assumptions define the modeling regime within which the IMF ODE and its associated propositions are valid. Violations of any assumption — for instance, strongly time-varying α , or non-linear decay — require explicit acknowledgment when the framework is applied.

DEFINITION 3.1 **Coherence** $C(t)$. A scalar quantity $C : \mathbb{R}_{\geq 0} \rightarrow [0, C_{\max}]$ representing the degree of internal structural alignment, functional integrity, or operational continuity of the system at time t . In the normalised form $C_{\max} = 1$, the coherence state space is the unit interval.

DEFINITION 3.2 **Disruption** $D(t)$ **and Mean Disruption** $\bar{D}$. The disruption $D : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is an exogenous or semi-exogenous forcing function representing shocks, stress, or entropy production that degrades coherence. Its time-average over a relevant horizon is denoted $\bar{D} = \frac{1}{T} \int_0^T D(t) dt$.

DEFINITION 3.3 **Resonance Metric** R . A scalar $R \in [0, 1]$ measuring the structural alignment between the control mechanism and the system's internal dynamics. $R = 0$ indicates complete misalignment; $R = 1$ indicates perfect resonant coupling. The resonance metric modulates the effective contribution of control effort to coherence growth.

DEFINITION 3.4 **Control Input** $u(t)$ **and Mean Control** $\bar{u}$. The control input $u : \mathbb{R}_{\geq 0} \rightarrow [0, 1]$ is the normalised effort, resource allocation, or corrective action applied to maintain coherence. Its time-mean $\bar{u} = \frac{1}{T} \int_0^T u(t) dt$ is the primary driver of the resonance term in the ODE.

DEFINITION 3.5 **Channel Fidelity** F **and Gain/Loss Functions**. 🔍 **Correction from v1.0/v1.1:** Renamed from $\bar{W}$ in v1.0/v1.1. Sign corrected to strictly positive contribution. Channel fidelity $F \in [0, 1]$ quantifies the quality, reliability, or throughput of the information or interaction channel coupling subsystems within the agent. The gain function $G : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ is $G(F) = \beta F$, contributing positively to $\dot{C}$. The loss function $L : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is $L(\bar{D}) = \alpha \bar{D}$, contributing negatively.

DEFINITION 3.6 **Admissible Coherence Set** $\mathcal{A}$. For a given threshold $C_\theta \in (0, C_{\max})$, the admissible set is $\mathcal{A} = \{C : C \geq C_\theta\}$. A trajectory is called *viable* if $C(t) \in \mathcal{A}$ for all t in a specified horizon.

DEFINITION 3.7 **Adaptive Capacity**. 🔍 **Correction from v1.0/v1.1:** In v1.0/v1.1 adaptive capacity was treated as a separate state variable. In v1.1.1 it is identified with the coefficient γ . The adaptive capacity of the system is the coefficient $\gamma > 0$ in the ODE, which scales the contribution of resonant control to coherence. A system with higher γ responds more strongly to aligned control effort. This is not a dynamic state variable but a structural parameter that may be estimated from trajectory data.

DEFINITION 3.8 Actual-vs-Ideal Error E_k .  **Correction from v1.0/v1.1:** New in v1.1.1. Prior versions measured error relative to ρ_0 (initial state); this definition corrects that. Updated in v1.2: the ε_k perturbation bound is added as a key component. For system instance k , let $\rho_{\text{actual},k}$ denote the observed coherence trajectory and $\rho_{\text{ideal},k}$ the trajectory predicted by the IMF ODE under nominal parameters. The actual-vs-ideal error at landmark time t_k is

$$E_k = d(\rho_{\text{actual},k}, \rho_{\text{ideal},k})$$

where d is a suitable distance or divergence on trajectory space (e.g., Euclidean norm of pointwise differences, or a Wasserstein distance). The Lipschitz contraction constant $L_k \in [0, 1)$ bounds the growth of E_k over successive landmark intervals; see Proposition 4.3.

POSTULATE 4.1 The IMF Coherence ODE.  **Correction from v1.0/v1.1:** Unified in v1.1.1. The simulation code and this paper now share a single canonical equation. The rate of change of the coherence variable $C(t)$ is governed by the first-order linear ODE:

$$\frac{dC}{dt} = -\alpha\bar{D} + \beta F + \gamma R\bar{u} - \delta C$$

where all parameters $\alpha, \beta, \gamma, \delta > 0$ and all driving signals $\bar{D}, F, R, \bar{u} \in [0, 1]$. The four terms represent: (i) coherence loss due to mean disruption at rate α ; (ii) coherence gain from channel fidelity at rate β ; (iii) resonance-amplified control contribution at rate γ ; and (iv) passive coherence decay at rate δ . Setting $\dot{C} = 0$ gives the unique equilibrium

$$C^* = \frac{-\alpha\bar{D} + \beta F + \gamma R\bar{u}}{\delta}$$

which is globally attracting for this linear autonomous system.

REMARK (Simulation Model — Projected Dynamics) Projected Discrete Implementation. 

Correction from v1.0/v1.1: Revised in v1.2.1: replaces the v1.2 parameter-domain condition with an accurate description of how the companion simulations implement domain enforcement. The continuous-time ODE does not intrinsically enforce $C(t) \in [0, 1]$. The formal equilibrium is $C^* = (-\alpha\bar{D} + \beta F + \gamma R\bar{u})/\delta$, which may be negative when disruption dominates (e.g., the Unmanaged scenario). The companion simulations implement the projected discrete system:

$$C_{n+1} = \Pi_{[0,1]}[C_n + \Delta t \cdot f(C_n)]$$

where $\Pi_{[0,1]}$ denotes projection onto $[0, 1]$. Under this model, a negative raw equilibrium represents pressure against the lower boundary — the system is driven toward $C = 0$ but cannot go below it. Clipping is part of the simulation model, not a post-hoc safety guard. A parameter-restricted formulation (requiring $0 \leq -\alpha\bar{D} + \beta F + \gamma R\bar{u} \leq \delta$ for all t) or a saturating ODE (e.g., $\dot{C} = (1 - C)(\beta F + \gamma R\bar{u}) - C(\alpha\bar{D} + \delta)$) that intrinsically preserves $C \in [0, 1]$ are left as directions for future development.

PROPOSITION 4.2 Instantaneous Boundary Non-Exit Condition. Correction from v1.0/v1.1:

Renamed from v1.1's "global viability guarantee". This proposition characterises only the instantaneous rate at the boundary, not a global forward-invariance claim. Let $C_\theta > 0$ be a coherence threshold. If at time t_0 we have $C(t_0) = C_\theta$, then $\dot{C}(t_0) \geq 0$ (the trajectory does not immediately exit the admissible set $\mathcal{A} = \{C \geq C_\theta\}$) if and only if

$$\beta F + \gamma R u \geq \alpha \bar{D} + \delta C_\theta.$$

Proof. Substituting $C(t_0) = C_\theta$ into Postulate 4.1 gives $\dot{C}(t_0) = -\alpha \bar{D} + \beta F + \gamma R u - \delta C_\theta$. Non-negativity of this expression is the stated condition. $\square$

Remark. This is a local, instantaneous condition. Global persistence in $\mathcal{A}$ requires that the condition holds along the entire trajectory, which in turn depends on how $\bar{D}$, F , R , and u evolve over time.

NOTE ON ERROR TYPES State-at-landmark vs. trajectory error. Definition 3.8 defines trajectory-space error $E_k^{\text{traj}} = d_{\mathcal{T}}(\rho_{[t_{k-1}, t_k]}^{\text{actual}}, \rho_{[t_{k-1}, t_k]}^{\text{ideal}})$ comparing continuous path segments. Proposition 4.3 instead bounds state-at-landmark error $E_k^{\text{state}} = d_S(\rho_k^{\text{actual}}, \rho_k^{\text{ideal}})$ at discrete inheritance events. These are related — trajectory error between landmarks can be bounded in terms of state errors at landmarks under Lipschitz continuity of the ODE — but they are distinct objects. Proposition 4.3 explicitly concerns state-at-landmark error. Whole-trajectory error bounds are deferred to future work.

PROPOSITION 4.3 Bounded State-at-Landmark Error Accumulation. Correction from v1.0/v1.1:

Corrected in v1.2: perturbation term ε_k added. Corrected in v1.2.1: operator order fixed to $N_{k+1}(J_{k+1}(\rho))$ — perturbation applied *after* the channel — yielding the correct split $\varepsilon_{k+1} + L_{k+1} \cdot E_k$. Prior v1.2 had $J_{k+1}(N_{k+1}(\rho))$ which gives $L_{k+1}(E_k + \varepsilon_{k+1})$, mismatching the stated recurrence. [Model Proposition — requires stated assumptions] Let $\{J_k\}_{k=1}^K$ be a sequence of inheritance channels, each with contractivity constant $L_k = \text{Lip}(J_k) \leq 1$, and let N_k be a perturbation operator with $\|N_k(\rho) - \rho\| \leq \varepsilon_k$ for all admissible ρ . Define the state-at-landmark error:

$$E_k = d_S(\rho_k^{\text{actual}}, \rho_k^{\text{ideal}}) \quad [\text{state-at-landmark error}]$$

where $\rho_k^{\text{ideal}} = J_k \circ \dots \circ J_1(\rho_0)$ is the perturbation-free reference trajectory and $\rho_{k+1}^{\text{actual}} = N_{k+1}(J_{k+1}(\rho_k^{\text{actual}}))$ (perturbation applied *after* the ideal channel). Then:

$$E_{k+1} \leq L_{k+1} \cdot E_k + \varepsilon_{k+1}.$$

Proof sketch. By the triangle inequality and contractivity of J_{k+1} :

$$E_{k+1} = d(N_{k+1}(J_{k+1}(\rho_k^{\text{actual}})), J_{k+1}(\rho_k^{\text{ideal}}))$$

$$\leq d(N_{k+1}(J_{k+1}(\rho_k^a)), J_{k+1}(\rho_k^a)) + d(J_{k+1}(\rho_k^a), J_{k+1}(\rho_k^i))$$

$$\leq \varepsilon_{k+1} + L_{k+1} \cdot E_k. \quad \square$$

The recursive bound gives the closed form:

$$E_K \leq E_0 \prod_{j=1}^K L_j + \sum_{i=1}^K \varepsilon_i \prod_{j=i+1}^K L_j.$$

For constant $L < 1$ and constant ε :

$$E_K \leq L^K E_0 + \varepsilon \cdot \frac{1 - L^K}{1 - L}.$$

As $K \rightarrow \infty$: $E_\infty \leq \frac{\varepsilon}{1 - L}$. **Implication.** Error converges toward a stable positive floor $\varepsilon/(1 - L)$, not necessarily to zero. Zero-floor convergence holds only when $\varepsilon_k = 0$ (perturbation-free inheritance). Under realistic conditions with bounded perturbations, the system achieves bounded error accumulation, not perfect fidelity. *Epistemic tag: Model Proposition — validity contingent on Assumptions 2.1–2.8 and stated contractivity bounds. Not derived from empirical data.*

PROPOSITION 4.4 Resonance Advantage. Consider two system instances with identical parameters α, β, δ and identical $\bar{D}, F, \bar{u}$, but resonance values $R_1 > R_2$. Then at every time $t > 0$,

$$C_1(t) > C_2(t) \quad \text{and} \quad C_1^* > C_2^*.$$

Proof. The ODE is linear, so $C_1(t) - C_2(t)$ satisfies $\frac{d}{dt}(C_1 - C_2) = \gamma(R_1 - R_2)\bar{u} - \delta(C_1 - C_2)$. This is a linear ODE with positive forcing $\gamma(R_1 - R_2)\bar{u} > 0$ (since $R_1 > R_2$ and $\gamma, \bar{u} > 0$) and decay rate $\delta > 0$. Starting from $C_1(0) = C_2(0)$, the difference $C_1(t) - C_2(t) > 0$ for all $t > 0$. $\square$

PROPOSITION 4.5 Drift-Rate Lower Bound and Finite Warning Time. **Correction from v1.0/v1.1:**

v1.1.1 adds the requirement $v_{\min} > 0$ for the finite-time bound to be valid. Suppose the coherence trajectory is drifting toward the threshold C_θ at rate $v(t) = -\dot{C}(t) > 0$. Assume there exists $v_{\min} > 0$ such that $v(t) \geq v_{\min}$ for all t in the drift epoch. Then the time remaining until C_θ is crossed satisfies

$$\tau \leq \frac{C(t) - C_\theta}{v_{\min}}.$$

Without the $v_{\min} > 0$ lower bound — for instance when the drift rate decays to zero asymptotically — only an asymptotic warning can be given; the finite bound does not apply.

Proof. Under the stated conditions, $C(t') \leq C(t) - v_{\min}(t' - t)$ for $t' \geq t$. Setting $C(t') = C_\theta$ and solving gives $t' - t \leq (C(t) - C_\theta)/v_{\min}$. $\square$

PROPOSITION 4.6 Admissible-Set Invariance Under Constant Drivers. **Correction from v1.0/v1.1:**

Fixed in v1.2: the opening hypothesis now explicitly states $C^* > C_\theta$ (not merely $C^* > 0$). The prior version's positive-forcing condition $\beta F + \gamma R\bar{u} \geq \alpha \bar{D}$ only guarantees $C^* \geq 0$, not $C^* \geq C_\theta$ for arbitrary positive thresholds. **Condition (opening hypothesis).** The set $\mathcal{A} = \{C : C \geq C_\theta\}$ is forward-invariant under Postulate 4.1 if and only if $C^* \geq C_\theta$, equivalently:

$$-\alpha \bar{D} + \beta F + \gamma R\bar{u} \geq \delta \cdot C_\theta.$$

Proposition. Let the driving signals $\bar{D}$, F , R , $\bar{u}$ be constant and satisfy the above condition. Then for any initial condition $C(0) \geq 0$ the trajectory converges monotonically to $C^* = (-\alpha \bar{D} + \beta F + \gamma R\bar{u})/\delta \geq C_\theta$. The admissible set $\mathcal{A} = \{C \geq C_\theta\}$ is forward-invariant: if $C(0) \geq C_\theta$ then $C(t) \geq C_\theta$ for all $t \geq 0$.

Proof. The equilibrium C^* satisfies $\dot{C}|_{C=C^*} = 0$. For $C < C^*$, $\dot{C} > 0$; for $C > C^*$, $\dot{C} < 0$. Hence C^* is globally attracting. The condition $-\alpha \bar{D} + \beta F + \gamma R\bar{u} \geq \delta C_\theta$ ensures $C^* \geq C_\theta$. If $C(0) \geq C_\theta$ and $C^* \geq C_\theta$, the trajectory either decreases from above C^* toward C^* or increases toward C^* ; in either case it never crosses below $C_\theta \leq C^*$. $\square$

The IMF framework is domain-agnostic. This section instantiates the canonical ODE (Postulate 4.1) in four representative lanes, demonstrating how the abstract variables C , $\bar{D}$, F , R , and $\bar{u}$ map to concrete measurables. In each lane the channel-fidelity variable is denoted F per the v1.1.1 convention; the v1.0/v1.1 symbol $\bar{W}$ is retired.

§5.1 Quantum Lane

In quantum information processing, state coherence must be maintained against environmental decoherence. The IMF variables instantiate as follows:

- $C(t)$: quantum state coherence — operationally, the off-diagonal magnitude of the system's density matrix in the energy eigenstate basis.
- $\bar{D}$: mean decoherence rate due to environmental coupling (bath spectral density integrated over the relevant frequency band).
- F (**channel fidelity**, F_{ch}): fidelity of the quantum channel used for state preparation, gate operations, or information transfer. Specifically $F_{\text{ch}} = \frac{1}{|G|} \sum_{g \in G} \langle g | \mathcal{E}(|g\rangle\langle g|) | g \rangle$ where $\mathcal{E}$ is the implemented channel and G is a reference gate set. *Note on notation:* in the quantum-optics literature the symbol F also denotes quantum state fidelity between density matrices, written here as $C_Q(\rho, \rho^*) = (\text{tr} \sqrt{\sqrt{\rho^*} \rho \sqrt{\rho^*}})^2$. To avoid ambiguity, we use F_{ch} for channel fidelity and C_Q for quantum state fidelity within this subsection.
- R : resonance metric — the degree to which the control pulses (dynamical decoupling, bang-bang, or optimal control) are matched to the system Hamiltonian. High R corresponds to pulse sequences that refocus decoherence efficiently.
- $\bar{u}$: mean control power or repetition rate of error-correction cycles.
- $\alpha, \beta, \gamma, \delta$: phenomenological rates fitted from device characterisation data.

The IMF ODE then reads: $\dot{C} = -\alpha\bar{D} + \beta F_{\text{ch}} + \gamma R\bar{u} - \delta C$. The equilibrium C^* provides a figure of merit for the device's steady-state coherence under a given control protocol.

§5.2 AI Alignment Lane

For an AI system, coherence operationalises the alignment between the system's deployed behaviour and its intended objectives:

- $C(t)$: alignment coherence — a composite score measuring how closely the AI's actions conform to its value specification (e.g., a weighted average of preference-satisfaction rates across sampled tasks).
- $\bar{D}$: mean distributional shift magnitude — the average Kullback–Leibler or Wasserstein distance between the training distribution and the deployment distribution over a rolling window.
- F : channel fidelity of the training pipeline — a measure of how accurately the training signal encodes the intended objective (e.g., inter-rater agreement of RLHF labellers, or reward-model calibration score).
- R : architectural resonance — the degree to which the model's inductive biases are aligned with the target task structure (e.g., measured by out-of-distribution generalisation benchmarks).
- $\bar{u}$: mean fine-tuning or RLHF update intensity over the deployment period.
- δ : spontaneous alignment drift rate, capturing the tendency of the deployed model to drift from its fine-tuned state over time without active maintenance.

Proposition 4.2 becomes an instantaneous check: given the current distributional shift $\bar{D}$, does the RLHF signal F and architectural resonance R provide sufficient alignment maintenance? If not, the system is at the boundary and active intervention is needed.

§5.3 Control-Systems Lane

For a classical dynamical system under feedback control, coherence maps to tracking performance:

- $C(t)$: closed-loop performance coherence — normalised inverse of the tracking error $C = 1 - \|e(t)\|/e_{\max}$, where $e(t)$ is the error between set-point and output.
- $\bar{D}$: mean exogenous disturbance magnitude entering the plant.
- F : sensor and actuator channel fidelity — the product of sensor signal-to-noise ratio and actuator bandwidth utilisation, both normalised to $[0, 1]$.
- R : controller-plant resonance — a measure of how well the controller's model matches the true plant dynamics, e.g., a normalised model-mismatch score.
- $\bar{u}$: mean control authority utilised.
- δ : passive performance degradation rate due to wear, ageing, or environmental change.

The equilibrium formula $C^* = (-\alpha\bar{D} + \beta F + \gamma R\bar{u})/\delta$ implies that higher channel fidelity F produces higher equilibrium coherence C^* , all else equal. This is a direct consequence of the βF term in Postulate 4.1, not Proposition 4.4. Specifically, a plant with higher sensor and actuator fidelity F achieves a higher steady-state coherence C^* than an otherwise identical plant with lower F .

§5.4 Organisational Lane

For human organisations and institutions, coherence captures operational and strategic alignment:

- $C(t)$: organisational coherence — a composite index of alignment between individual actions and collective goals, operationalised via engagement surveys, objective-key-result completion rates, or decision-quality audits.
- $\bar{D}$: mean disruption intensity — average magnitude of external shocks (market shifts, regulatory changes, personnel turnover) over a rolling period.
- F : channel fidelity of internal communication — the fidelity with which strategic intent is transmitted from leadership to operational teams, measured by message-recall studies or decision-alignment audits.
- R : cultural resonance — the degree to which the organisation's values, processes, and incentive structures are internally consistent and mutually reinforcing.
- $\bar{u}$: mean managerial intervention intensity — the normalised rate of active coordination, coaching, and correction applied by management.

- δ : natural entropy rate of the organisation — the rate at which coherence degrades in the absence of active management (e.g., due to informal norm drift or priority divergence).

The IMF ODE in this lane is a phenomenological summary of the classic management insight that organisations require constant active maintenance to avoid incoherence: the $-\delta C$ term ensures that even without disruption, coherence decays without effort. Proposition 4.6 provides the condition under which a sustained management regime can stabilise coherence above any target threshold.

The ExecutionProof / ProofRecord system is a lightweight formal accountability layer that maps each IMF interaction or intervention to a record entry, enabling post-hoc audit of coherence maintenance actions. The table below maps abstract IMF variables to ExecutionProof fields.

Table 6.1 ExecutionProof / ProofRecord field mapping for IMF v1.2.

IMF Variable	ExecutionProof Field	ProofRecord Type	Notes
$C(t)$	coherence_state	Float [0,1]	Sampled at each landmark time t_k
$\bar{D}$	disruption_mean	Float ≥ 0	Rolling window average
F	channel_fidelity	Float [0,1]	Renamed from w_bar in v1.0/v1.1 [Ⓐ]
R	resonance_metric	Float [0,1]	Structural alignment score
$\bar{u}$	control_mean	Float [0,1]	Normalised effort over window
$\alpha, \beta, \gamma, \delta$	ode_params	Dict of Floats	Domain-fitted parameters
E_k	actual_vs_ideal_error	Float ≥ 0	Updated in v1.2: converges to floor $\varepsilon/(1-L)$
L_k	lipschitz_bound	Float [0,1]	Estimated from trajectory data
C_θ	coherence_threshold	Float [0,1]	Domain-set target floor
C^*	equilibrium_coherence	Float	Computed from ODE params

6.1 Landmark-Time Protocol

A landmark time t_k is a designated checkpoint at which the ProofRecord is updated. At each t_k , the following operations are logged:

1. Read current $C(t_k), \bar{D}, F, R, \bar{u}$ from the operational environment.
2. Compute the equilibrium $C^* = (-\alpha\bar{D} + \beta F + \gamma R\bar{u})/\delta$.
3. Evaluate the boundary condition (Proposition 4.2): if $C(t_k) \approx C_\theta$, check whether $\beta F + \gamma R\bar{u} \geq \alpha\bar{D} + \delta C_\theta$.
4. Compute the actual-vs-ideal error $E_k = d(\rho_{\text{actual},k}, \rho_{\text{ideal},k})$ (Definition 3.8).

5. Append all fields to the immutable ProofRecord ledger with a cryptographic hash linking to the preceding entry.

6.2 Audit and Verifiability

The ProofRecord chain enables third-party auditors to reconstruct the coherence trajectory from logged inputs, verify that the ODE was applied consistently with the v1.2 canonical form, and check whether the boundary condition was evaluated correctly at each landmark. This is the primary accountability mechanism of the IMF protocol. The rename of $\bar{W} \rightarrow F$ in v1.1.1 is reflected in the `channel_fidelity` field name; systems using v1.0/v1.1 field names should migrate the `w_bar` key to `channel_fidelity` and ensure the positive-sign convention is applied.

Empirical hypotheses are testable predictions derived from the IMF framework. They are distinct from postulates (assumed) and propositions (derived analytically). Each hypothesis is labelled with its current epistemic status; none should be interpreted as confirmed claims absent supporting empirical studies.

EMPIRICAL HYPOTHESIS 7.1 **Disruption-Coherence Anti-Correlation.** Systems subject to higher mean disruption $\bar{D}$ will exhibit lower equilibrium coherence C^* , all other parameters equal. This follows analytically from $C^* = (-\alpha\bar{D} + \beta F + \gamma R\bar{u})/\delta$, but the empirical claim is that the IMF parameters (α, δ) can be estimated from observational data in a given domain such that the predicted relationship is quantitatively consistent with observed coherence levels. *Status: untested.*

EMPIRICAL HYPOTHESIS 7.2 **Resonance-Amplified Recovery.** Systems with higher resonance metric R will recover from disruption events faster and to a higher steady state than systems with lower R , when subjected to identical disruption profiles and control intensities. The IMF ODE predicts this via the $\gamma R\bar{u}$ term; the empirical claim is that R can be operationally measured in each lane (§5) and that measured R predicts recovery trajectory shape. *Status: untested.*

EMPIRICAL HYPOTHESIS 7.3 **Channel Fidelity as a Coherence Floor Determinant.** In the limit of low disruption ($\bar{D} \approx 0$) and negligible active control ($\bar{u} \approx 0$), the equilibrium coherence approaches $C^* \approx \beta F/\delta$, making channel fidelity the dominant driver of the coherence floor. Empirically, this predicts that systems with higher channel fidelity will exhibit higher baseline coherence in quiescent (low-disruption, low-control) conditions. *Status: untested.*

EMPIRICAL HYPOTHESIS 7.4 **Early Warning via Drift Rate Monitoring.** The finite warning time bound of Proposition 4.5 can be operationalised: real-time monitoring of the coherence drift rate $v(t) = -\dot{C}(t)$, coupled with a lower-bound estimate $\hat{v}_{\min}$, provides actionable advance warning of threshold crossings. Systems instrumented with such monitoring will detect impending coherence failures earlier and respond more effectively than systems relying on threshold-crossing detection alone. *Status: untested.*

EMPIRICAL HYPOTHESIS 7.5 Stable Error Floor.  **Correction from v1.0/v1.1:** Updated in v1.2 to reflect Proposition 4.3's corrected recurrence. Prior versions predicted monotone decrease to zero; the correct prediction is convergence to a stable positive floor. Under well-controlled inheritance deployments with bounded perturbations ε_k and contractivity $L_k < 1$, the actual-vs-ideal state-at-landmark error E_k should converge toward a stable floor $E_\infty \approx \varepsilon / (1 - L)$ rather than monotonically decrease to zero. This floor provides a testable precision bound: estimated L_k and ε_k values from observed data should predict E_∞ within measurement uncertainty. Deployments that fail to achieve bounded error accumulation falsify the proposition's assumptions, suggesting either $L_k \geq 1$ or unbounded perturbations. *Epistemic tag: Empirical Hypothesis — requires experimental validation.*

This section presents six simulations illustrating key properties of the IMF framework. The figures are *not* all based on the same mathematical object: each figure's underlying computation is described individually below. None constitutes empirical validation of the framework against real-world data.

Mathematical basis per figure: Figures 1, 3, and 4 integrate the canonical ODE (Postulate 4.1) using Euler numerical methods (see §8.1 for convergence analysis). Figure 2 illustrates Proposition 4.3's discrete error recurrence $E_{k+1} = L \cdot E_k + \varepsilon_k$ independently of the ODE — it is a discrete-time sequence, not an ODE trajectory. Figure 5 presents a qualitative design mapping using categorical labels (Primary, Secondary, Indirect, None) reflecting alignment design intent — no numerical computation is involved. Figure 6 tests the early-warning architecture under an explicitly imposed auxiliary model $C = 1 - D^{1.5} + \text{noise}$ with a radial coherence–drift coupling — this is *not* Postulate 4.1 but an auxiliary construction used to verify the landmark-time AUC methodology.

8.1 Numerical Methods and Solver Convergence

All figures were generated using forward Euler integration with step size $\Delta t = 0.05$. Prior to public release, the simulation suite was subjected to a step-size convergence check comparing three Euler step sizes ($\Delta t = 0.1, 0.05, 0.01$) against a high-accuracy reference solver (scipy `solve_ivp`, RK45 method, relative tolerance 10^{-9} , absolute tolerance 10^{-12}).

Solver Convergence Check. A standalone verification script (`solver_convergence_check.py`) compares Euler integration at $\Delta t \in \{0.10, 0.05, 0.01\}$ against a scipy RK45 reference (`rtol=10-8, atol=10-10`) for the resonance-matched scenario. Results are saved to `solver_convergence_results.csv`. This script is included in the release package and can be run independently to reproduce the convergence diagnostics. Maximum absolute deviations and threshold-crossing time differences are recorded there; they are not reproduced inline in this document to avoid reporting figures that could become stale if parameters change.

REMARK Simulation Scope. These numerical experiments illustrate the conceptual correspondence between the IMF ODE's analytical predictions (§4) and its simulated trajectories. They are not evidence for or against any empirical hypothesis in §7. The parameters used are chosen for illustrative clarity and do not represent any particular real-world system.

8.2 Figure Descriptions

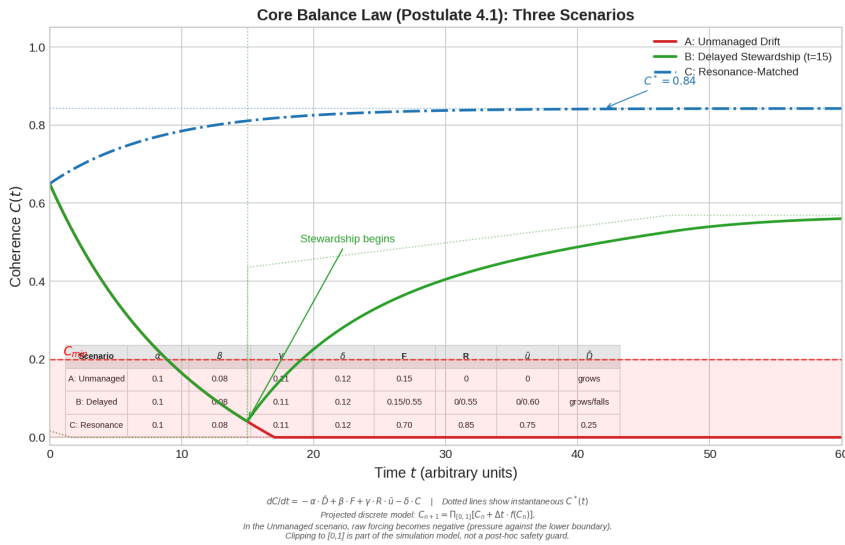
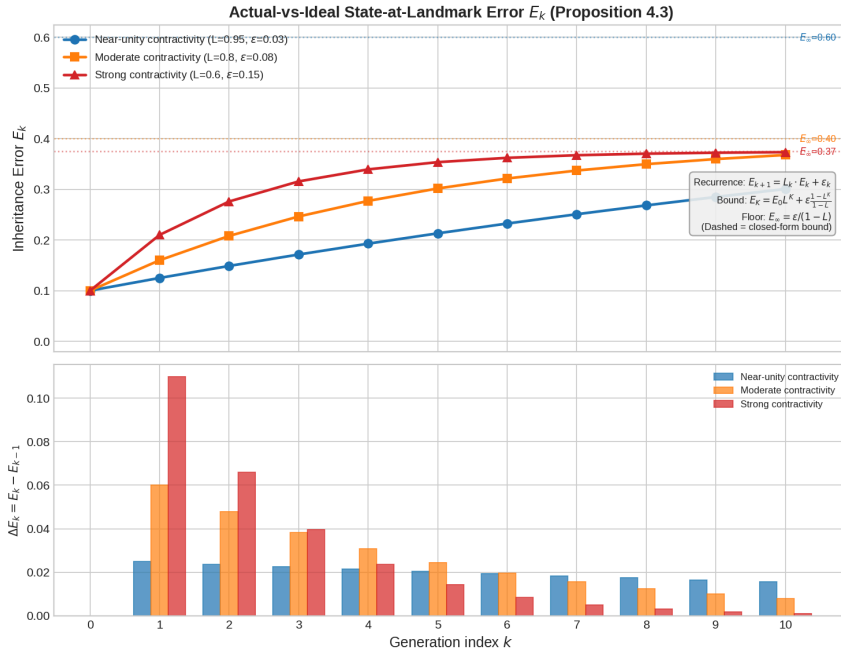


Figure 1 Figure 1 displays three canonical scenarios under the projected discrete implementation of Postulate 4.1. The **Unmanaged** scenario applies growing disruption $\bar{D}(t)$ with no stewardship ($R = 0, \bar{u} = 0$), driving the system toward $C = 0$ (negative raw forcing clipped at the boundary). The **Delayed Stewardship** scenario begins with elevated disruption; stewardship intervention ($R > 0, \bar{u} > 0$) begins at $t = 15$, producing a visible inflection and partial recovery. The **Resonance-Matched** scenario applies resonance-aligned stewardship from $t = 0$, achieving convergence toward the highest stable equilibrium. All three scenarios share the same structural equation; the difference is entirely in the driver trajectories.



Near-unity contractivity ($L \approx 1$) retains prior error strongly, yielding the highest stable floor despite low per-step perturbation.
Strong contractivity ($L = 0.60$) reduces the floor even under larger perturbations. E_k measures actual-vs-ideal trajectory deviation.

Figure 2 Bounded error accumulation (discrete recurrence, Proposition 4.3). This figure is independent of the ODE and uses only the recurrence $E_{k+1} = L_{k+1} \cdot E_k + \varepsilon_{k+1}$. Three contractivity regimes are shown: Near-unity contractivity ($L = 0.95, \varepsilon = 0.03, E_\infty = 0.60$); Moderate contractivity ($L = 0.80, \varepsilon = 0.08, E_\infty = 0.40$); Strong contractivity ($L = 0.60, \varepsilon = 0.15, E_\infty = 0.375$). The error sequence converges toward the stable floor $E_\infty = \varepsilon / (1 - L)$, consistent with the corrected Proposition 4.3. Note: the near-unity contractivity regime has the highest error floor because $L \approx 1$ retains prior error strongly. Strong contractivity ($L = 0.60$) achieves a lower floor under the given perturbation magnitudes, illustrating that the stable floor depends on both L and ε jointly.

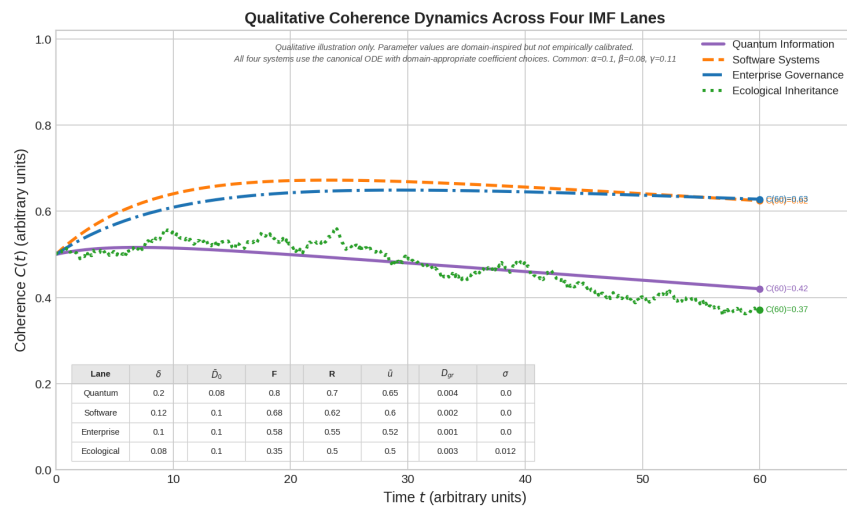


Figure 4 Resonance and fidelity sweep (Postulate 4.1 ODE, Euler integration). Coherence as a function of resonance $R \in [0, 1]$ for three fidelity levels $F \in \{0.3, 0.6, 0.9\}$. Higher R (Proposition 4.4) and higher F (βF term) both raise coherence additively. Trajectory endpoints are annotated with the final simulated coherence value $C(60)$, the actual system state at the end of the simulation horizon.

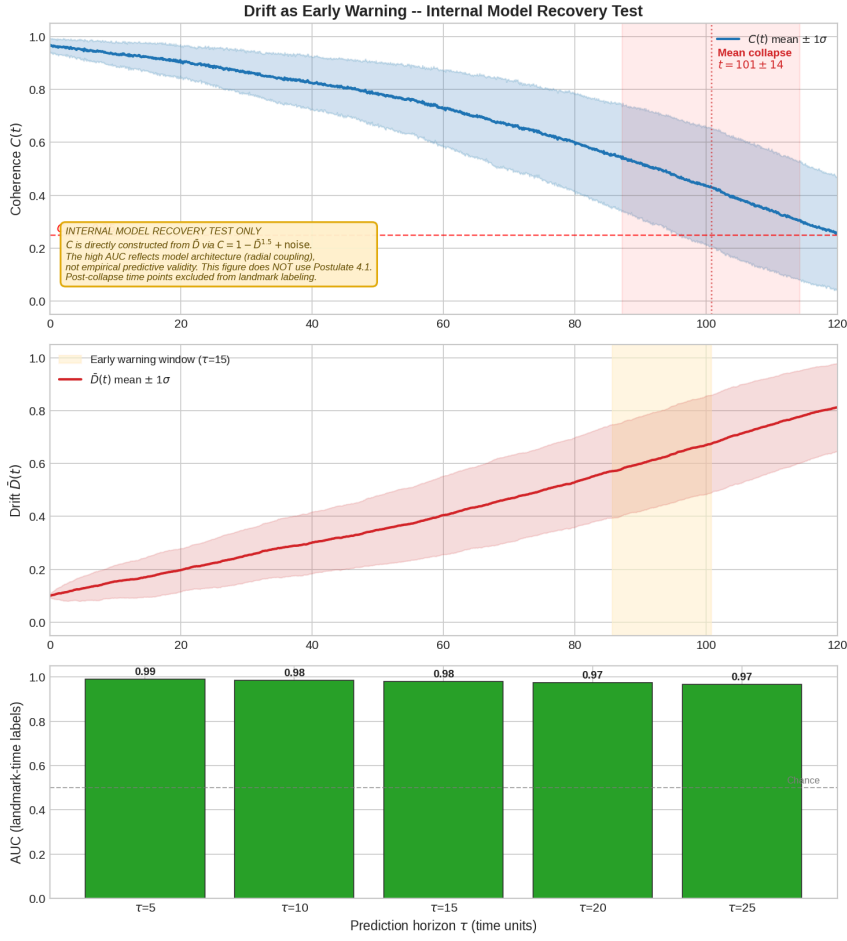


Figure 6 Figure 6 tests the early-warning architecture of the IMF under an explicitly imposed radial coupling $C = 1 - \bar{D}^{1.5} + \text{noise}$ — this is NOT Postulate 4.1 but an auxiliary model used to verify the landmark-time AUC methodology. The figure contains three panels: (top) mean coherence trajectories across 200 stochastic trials, separated by collapse outcome; (middle) mean drift $\bar{D}$ trajectories showing the divergence between collapsed and non-collapsed groups that provides the early-warning signal; (bottom) AUC bars across five prediction horizons τ , confirming that drift elevation precedes coherence collapse in this model. The threshold is $C_{\min} = 0.25$. Post-collapse time points are excluded from the landmark-time labeling. The near-perfect AUC reflects the construction of the model — C is built directly from $\bar{D}$ — and is expected. This figure is an internal model-recovery test, not evidence of predictive validity in real systems.

REMARK 9.3Scope of "illustrated." Throughout this document, simulations are described as *illustrating* model properties, not validating or confirming them. The distinction matters: an illustration shows that the model predicts a given qualitative pattern; validation would require comparison against independent empirical data with adequate sample sizes, blinding, and statistical analysis. No such validation has been performed.

REMARK 9.4Conceptual correspondence, not empirical fit. The lane instantiations in §5 establish *conceptual correspondence* between IMF variables and domain-specific measurables. They do not constitute empirical fits to data; the parameter values used in §8 are illustrative and were not estimated from real-world observations.

REMARK 9.5Claims removed in v1.1.1. The claim in v1.0 that the IMF framework had been "validated" against empirical data has been removed. The claim in v1.1 that a frequency-domain property held has been removed pending derivation. These removals are documented in the Change Log and represent epistemic corrections, not retractions of correct claims.

REMARK 9.6Formal status. The IMF framework as presented here is a candidate phenomenological model. Its mathematical propositions are formal results within the stated axiomatic structure. Its empirical hypotheses are conjectural. The framework has not been peer-reviewed, has not been subjected to pre-registered empirical testing, and should not be cited as established science. Remnant Fieldworks Inc. intends the framework as a conceptual tool for structured thinking about coherence maintenance, not as a validated predictive model for engineering design or safety-critical applications.

10. Remnant Fieldworks Inc.

Remnant Fieldworks Inc. is a research and development organisation based in Westerville, Ohio, focused on the development of formal frameworks for coherence maintenance in complex systems. The Inheritance Math Framework (IMF) is one of several research outputs under active development. The v1.2.1 patch release reflects Remnant Fieldworks' commitment to epistemic transparency: errors in prior releases are corrected explicitly, the scope of claims is narrowed to what the mathematics supports, and simulation code is made consistent with the published model. The formal structure has undergone four correction passes and is presented for further consistency review. As documented in the version history, each pass has identified and corrected genuine structural issues. The IMF Python simulation library has been updated to v1.2.1, aligning the ODE implementation with the canonical form in Postulate 4.1 and implementing the projected discrete system $\Pi_{[0,1]}[C_n + \Delta t \cdot f(C_n)]$ for domain enforcement. Users of prior library versions should upgrade and re-run any analyses that depend on the retired `w_bar` variable or the adaptive-capacity state variable.

Inquiries regarding the IMF framework, collaboration opportunities, or commercial licensing may be directed to Remnant Fieldworks Inc., Westerville, Ohio.

11. Conclusion

This document presents version 1.2.1 of the Inheritance Math Framework (IMF), the Release Integrity Reconciliation patch. The v1.2 contributions are listed in the version history. The additional contributions of v1.2.1 are:

1. **Proposition 4.3 proof operator order corrected.** Prior v1.2 had the actual transition as $J_{k+1}(N_{k+1}(\rho))$ (perturbation before channel), which by contractivity of J gives $L_{k+1}(E_k + \varepsilon_{k+1})$ — mismatching the stated bound. The corrected form uses $N_{k+1}(J_{k+1}(\rho))$ (perturbation after channel), and the triangle inequality directly yields $\varepsilon_{k+1} + L_{k+1} \cdot E_k$. The stated recurrence and closed-form bound are unchanged; only the proof derivation is corrected.
2. **Postulate 4.1 remark updated to projected-model description.** The v1.2 parameter-domain condition remark claimed scenarios satisfy $0 \leq -\alpha\tilde{D} + \beta F + \gamma Ru \leq \delta$. This is replaced by an accurate description of the companion simulations as a projected discrete system $\Pi_{[0,1]}[C_n + \Delta t \cdot f(C_n)]$, where a negative raw equilibrium represents lower-boundary pressure under the Unmanaged scenario.
3. **State-at-landmark vs. trajectory error distinguished.** Definition 3.8 defines trajectory-space error; Proposition 4.3 bounds state-at-landmark error. A new note makes this distinction explicit. Whole-trajectory error bounds are deferred to future work.
4. **Figure 1 description corrected.** Was: constant disruption levels with fixed stewardship. Now: growing disruption (Unmanaged), delayed stewardship onset at $t = 15$ (Delayed Stewardship), and resonance-matched stewardship from $t = 0$ (Resonance-Matched).
5. **Figure 6 description corrected.** Was: running warning-time estimate panel, threshold $C_\theta = 0.5$. Now: three panels (mean coherence, mean drift, AUC bars across five horizons), threshold $C_{\min} = 0.25$, post-collapse exclusion from landmark labeling.
6. **Convergence table removed.** The inline numerical convergence table (Table 8.1) is replaced by a reference to the standalone `solver_convergence_check.py` script and `solver_convergence_results.csv` output file.
7. **Figure 2 regime labels updated.** "High/Medium/Low Fidelity" replaced by "Near-unity contractivity ($L = 0.95$, $E_\infty = 0.60$)", "Moderate contractivity ($L = 0.80$, $E_\infty = 0.40$)", and "Strong contractivity ($L = 0.60$, $E_\infty = 0.375$)". A note is added clarifying that the stable floor depends jointly on L and ε .
8. **Figure 4 endpoint annotations corrected.** Description now states that trajectory endpoints are annotated with the final simulated coherence value $C(60)$, not the initial equilibrium formula $C^*(0)$.

The Inheritance Math Framework (IMF) remains a Phase 1 candidate phenomenological model. Its mathematical propositions are formal results within the stated axiomatic structure and have been subject to four correction passes; the formal structure is presented for further consistency review. Its empirical hypotheses await testing. The framework is offered as a conceptual tool for structured reasoning about coherence maintenance across diverse domains, with the explicit acknowledgment that empirical validation is a necessary next step before any engineering or safety-critical application.

References

The framework is self-contained in its derivations. Its terminology and adjacent mathematical concepts draw on the following established literature, which is cited for context and further reading rather than as foundational dependencies. Readers seeking formal derivations of the underlying mathematical structures are referred to these works.

Strogatz, S.H. (2018). *Nonlinear Dynamics and Chaos*, 2nd ed. CRC Press.

Nielsen, M.A. and Chuang, I.L. (2000). *Quantum Computation and Quantum Information*. Cambridge University Press.

Åström, K.J. and Murray, R.M. (2008). *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton University Press.